

CDS-492: Capstone for Data Science
Fall 2025
Project Presentation

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Topic

Question: Can online fan engagement measured through Google Trends predict next-week MLB stadium attendance?

MLB fans interact digitally (search engines, news spikes, roster updates). Does this digital attention translate into real in-person turnout?

Hypothesis: Rising online attention in week $t - 1$ will be followed by higher attendance in week t .

Significance

- **Sports Business:** Attendance forecasting affects staffing, promotions, and ticket pricing.
- **Data Science:** Tests whether digital traces can predict real-world human behavior.
- **Sports Analytics:** Incorporates high-frequency digital signals beyond traditional predictors.

Prior Research

- Traditional MLB attendance models emphasize win rate, market size, stadium effects.
- Google Trends provides real-time information about public attention.
- Gap: Few studies test whether weekly search activity predicts stadium attendance.
- This project evaluates that missing link.

Deliverables

- Capstone project paper (full econometric analysis)
- Science article (IEEE format)
- Final slide presentation
- Reproducible R pipeline (panel dataset + FE regression)

Method of Inquiry

- Estimate log-linear two-way fixed effects regression
- Collect MLB attendance logs (Baseball-Reference)
- Gather Google Trends daily search indexes
- Convert to weekly team-level dataset
- Apply lag structure: $\text{Trends}_{t-1} \rightarrow \text{Attendance}_t$

Theory

$$\ln(Attendance_{i,t}) = \beta_1 \ln(Trend_{i,t-1}) + \beta_2 WinRate_{i,t-1} + \beta_3 HomeGames_{i,t} +$$

- Team fixed effects remove market-size differences.
- Week fixed effects remove seasonality and league-wide shocks.
- Tests whether digital interest contains predictive information.

Results

Regression Estimates

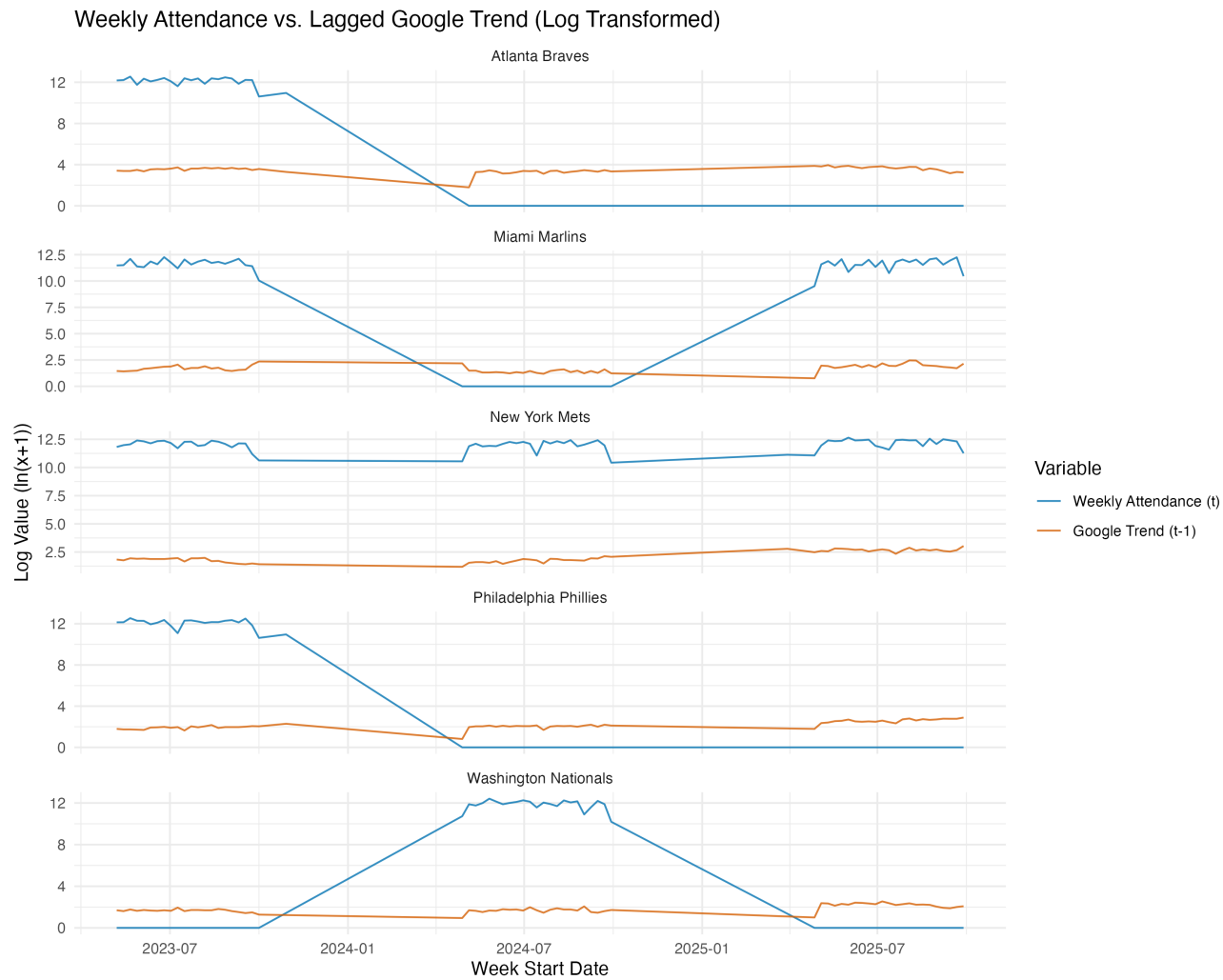
$$\beta_1(\text{Lagged Trends}) = -1.60, \quad p = 0.00027$$

$$\beta_2(\text{Win Rate}) = 28.9, \quad p < 2 \times 10^{-16}$$

- Digital search spikes → **lower next-week attendance.**
- Higher win rate → **higher attendance.**
- HomeGames: not significant once FE controls applied.

Conclusion: Reject H_1 . Google Trends does not increase attendance.

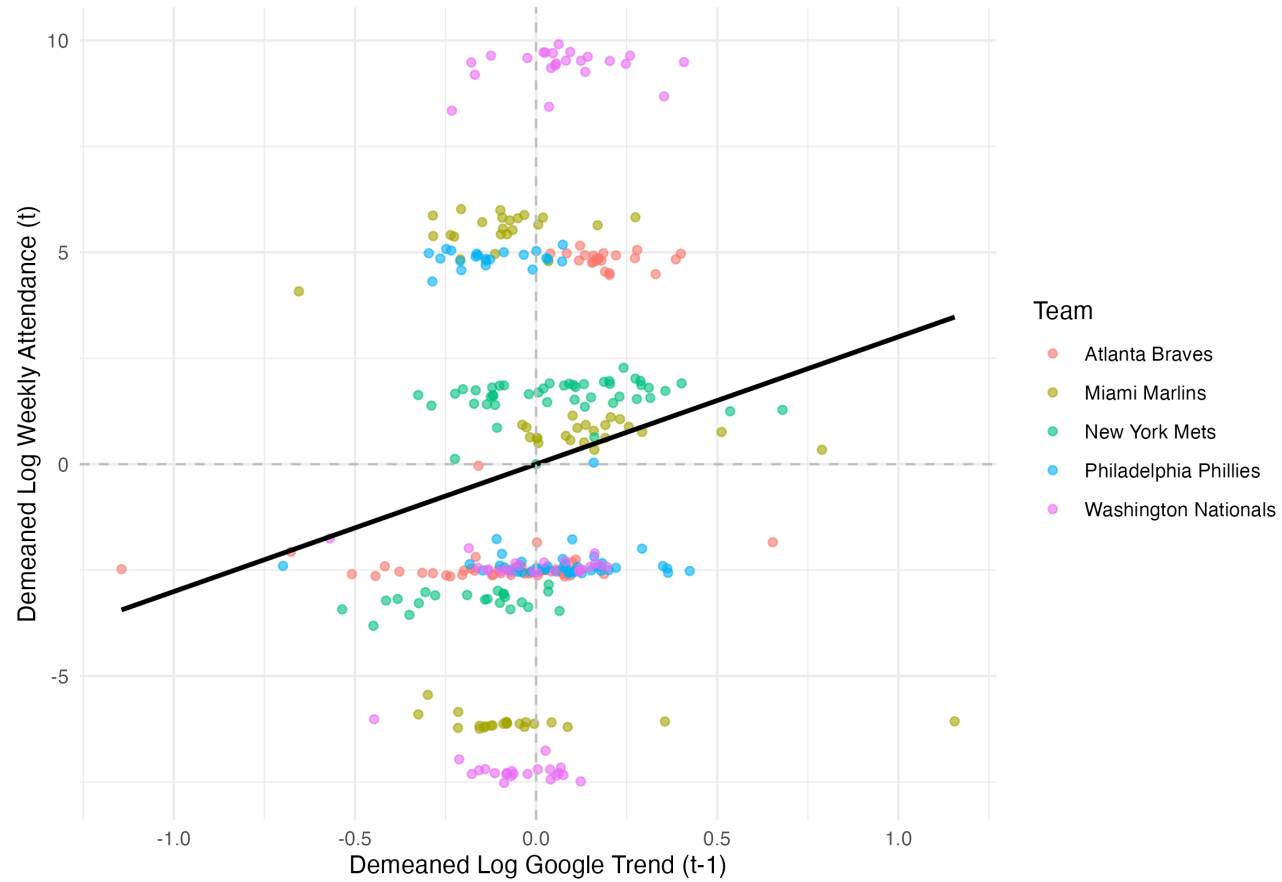
Results: Visualization



Search spikes often occur shortly before dips in stadium attendance.

Results: Fixed Effects Relationship

Within-Relationship: Log Attendance vs. Lagged Log Google Trend
Both variables are Demeaned (Team and Time Fixed Effects Removed)



Within-team variation shows a clear negative slope: higher search → lower following-week attendance.

Conclusion

- Google Trends is **not** a positive leading indicator of MLB attendance.
- Search spikes reflect curiosity, news, or uncertainty — not turnout.
- Win rate remains the strongest predictor of fan demand.
- Future extensions: sentiment filtering, event tagging, multi-platform attention.

Source Materials

- Baseball-Reference (attendance logs)
- Google Trends search index
- R scripts (panel construction + FE regression)
- mathematics.bib, science.bib (20+ references)